

Preparing for Influenza Season: Interim Report

1. Project Overview

Motivation: The United States has an influenza season where more people than usual suffer from the flu. Some people, particularly those in vulnerable populations, develop serious complications and end up in the hospital. Hospitals and clinics need additional staff to adequately treat these extra patients. The medical staffing agency provides temporary staff.

Objective: Determine when to send staff, and how many, to each state.

Scope: The agency covers all hospitals in each of the 50 states of the United States, and the project will plan for the upcoming influenza season. [Business Requirements in Appendix A.](#)

2. Hypothesis

If the size of the vulnerable population (patients are older than 65 years old) increases, then the risk of mortality increases. [Hypothesis Development in Appendix B.](#)

3. Data Overview

Influenza Deaths Data Set: This data contains monthly death counts for influenza-related deaths in the United States from 2009 to 2017. This data is categorized by state and age group.

Population Data Set: This data shows the population distribution by age and gender across the United States from 2009 to 2017. [Data Set Profiles in Appendix C and D.](#)

4. Data Limitations

Influenza Deaths Data Set

Suppressed Data: Due to privacy concerns, 82% of death count data was suppressed. Missing values were imputed, potentially affecting analysis accuracy.

Missing Age Data: 8.33% of records had age listed as “Not Stated”. These were included in “All” counts but excluded in age-specific counts, potentially affecting trend accuracy by age group.

Population Data Set

Missing values: Some counties lacked population estimates for certain years, potentially leading to inaccuracies in state-level death rate analysis.

Duplicate values: A significant portion of the data had County-Year duplicates with discrepancies in population values. These were removed randomly, potentially affecting the accuracy of total population estimates.

5. Descriptive Analysis

US Census				Influenza Deaths		
	(0-64 years)	(65+ years)	total population	(0-64 years)	(65+ years)	total deaths
Mean	5167038	806989	5973849	490	890	1381
Standard Deviation	5938164	886050	6799372	124	976	1087

The size of the vulnerable population shows a strong correlation (0.9) with influenza-related deaths, indicating that **states with a larger elderly population tend to have higher death rates**.

The standard deviation being relatively higher than the mean, shows that **deaths vary widely between states**.

When comparing the mean of deaths from the 65+ population and the mean of total deaths, we can see that the **individuals aged 65 and older account for 64% of total deaths (890/1381)**, emphasizing their significant contribution to overall mortality.

6. Results and Insights

Statistical Hypothesis

Null hypothesis: The size of the vulnerable population (65+) has no effect on the risk of mortality.

Alternative hypothesis: The size of the vulnerable population (65+ years) has a positive effect (increases) on the risk of mortality.

At a 95% confidence level ($\alpha=0.05$) **the size of the vulnerable population (65+) has a significant positive effect on mortality risk** across states. This insight supports the project's goal, as it helps develop a staffing plan for the upcoming influenza season, given that states with larger elderly populations face a higher risk of hospitalization. [Statistical Test in Appendix E](#).

7. Remaining Analysis and Next Steps

Additional Analysis:

- Correlation Analysis: Explore the relationship between other vulnerable populations (e. g., pregnant women, children under 5) and mortality risk.
- Age Group Comparison: Examine correlations between non-vulnerable population and mortality risk to identify broader trends.
- Comparison Charts: Display mortality rates across different populations.
- Temporal Analysis: create time-series visualizations to forecast trends for the upcoming influenza season.
- Statistical Visualizations: Illustrate variable distributions and correlations to identify key patterns and relationships.

- Heat, Density or Choropleth Map: visualize influenza-related deaths by region to identify geographical patterns and high-risk areas.
- Word Cloud: Generate a visual representation of qualitative data to highlight key terms and recurring themes related to influenza.
- Tableau Storyboard: Publish an interactive analysis to present findings visually.
- Video presentation: Create a concise and engaging summary for stakeholders.

8. Appendix

- a. Business Requirements
- b. Hypothesis Development

Key Questions:

1. Which flu cases result in hospitalizations?

Hospitalizations and death due to influenza occur mainly among high-risk groups.

2. Are vulnerable population more likely to be hospitalized due to the flu?

Yes, pregnant women, children under 5 years of age, older people (65 years and older), individuals with chronic illness (such as chronic cardiac, pulmonary, renal, metabolic, neurodevelopmental, liver or hematologic diseases) and individuals with immunosuppressive conditions/treatments (such as HIV, receiving chemotherapy or steroids, or malignancy) are more at risk than others. But also, health and care workers due to increased exposure to the patients, and of further spreading particularly to vulnerable individuals.

Sources Consulted:

- World Health Organization. (n.d.). *Influenza (seasonal)*. Retrieved January 8, 2025, from [https://www.who.int/news-room/fact-sheets/detail/influenza-\(seasonal\)](https://www.who.int/news-room/fact-sheets/detail/influenza-(seasonal))
- Hampson, A. W. (2013). The influenza viruses. *Australian and New Zealand Journal of Public Health*, 37(2), 95-96. <https://doi.org/10.1111/1753-6405.12170>

- c. Influenza Deaths Data Set Profile
- d. Population Data Set Profile
- e. Statistical Test